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(54) **PORTABLE THREE-DIMENSIONAL IMAGE
MEASURING DEVICE,
THREE-DIMENSIONAL IMAGE
MEASURING METHOD USING SAME, AND
MEDICAL IMAGE MATCHING SYSTEM**

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014106 on Oct. 13, 2021, now Pat. No. 12,318,188.

(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

A portable three-dimensional image measuring device according to various embodiments of the present disclosure may include a light source configured to output patterned light; a camera configured to generate a light field image of an object by receiving reflected light generated by reflecting the patterned light from the object; and an optical path control element configured to reflect the patterned light output from the light source so that the object is irradiated with the patterned light, and to transmit the reflected light reflected from the object so that the reflected light reaches the camera. An optical path of the patterned light output from the light source and irradiated onto the object and an optical path of the reflected light reflected from the object and reaching the camera may overlap coaxially in a section between the optical path control element and the object.

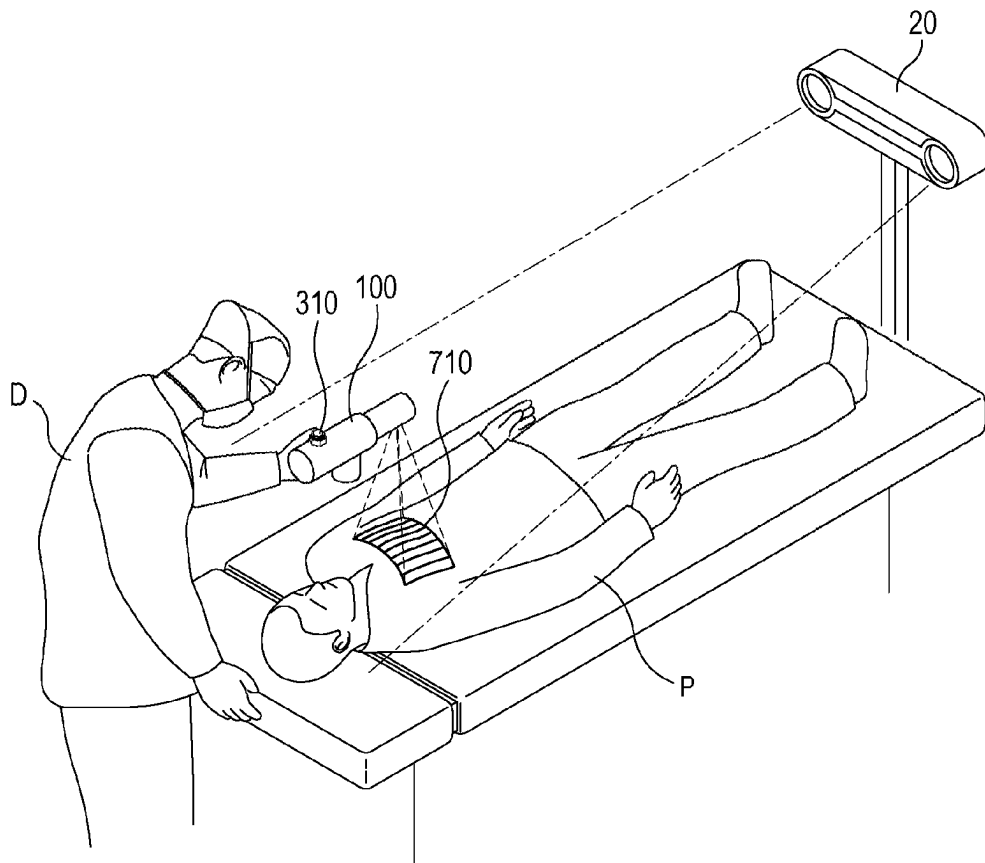


FIG. 1

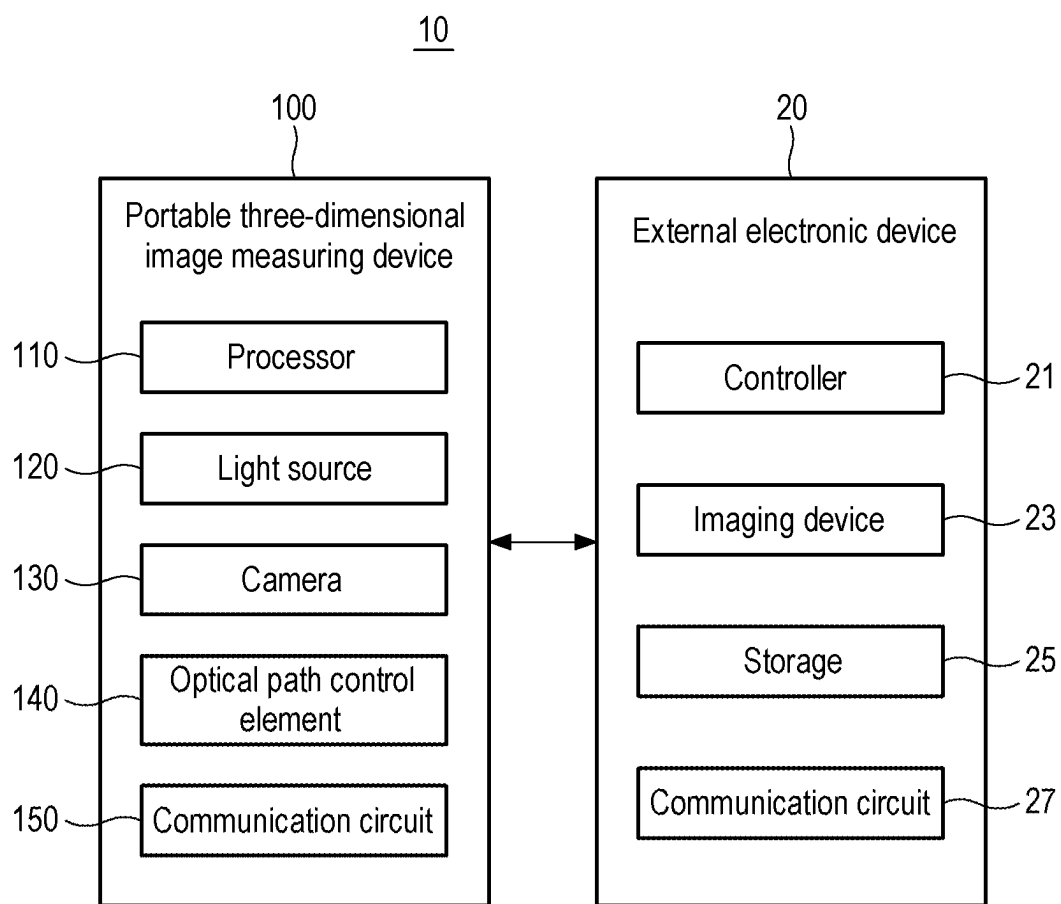


FIG. 2A

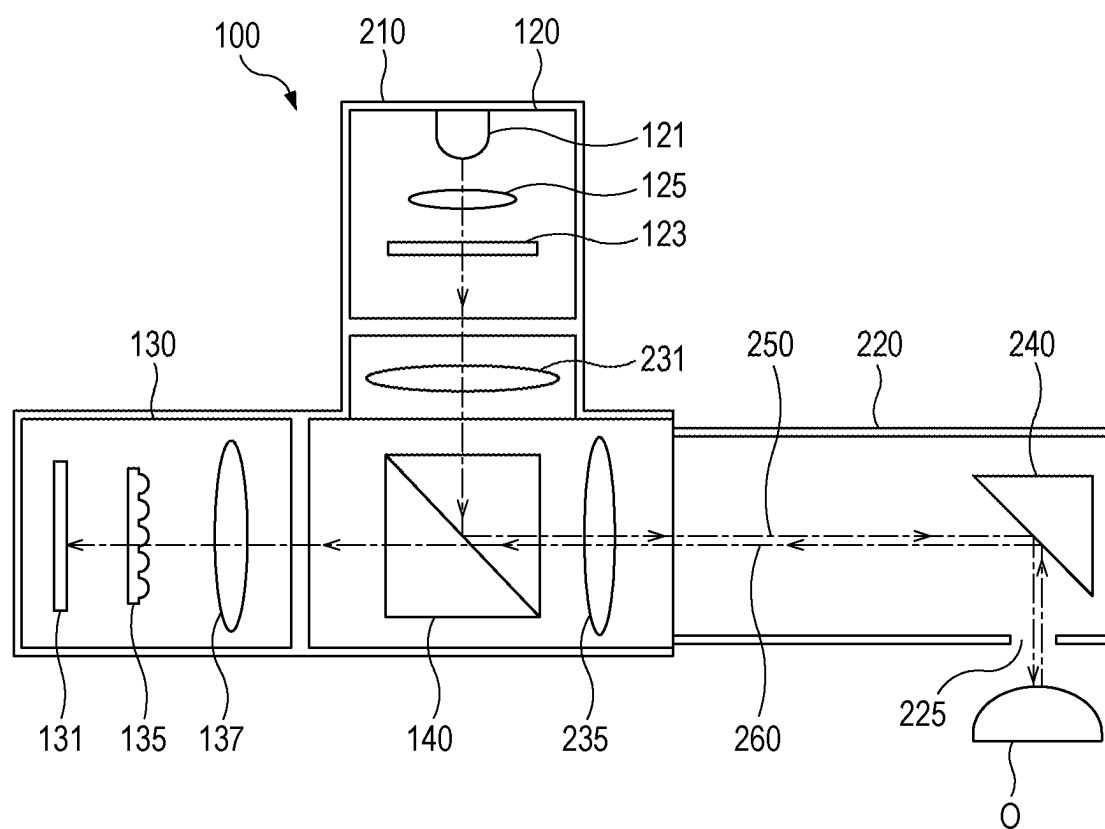


FIG. 2B

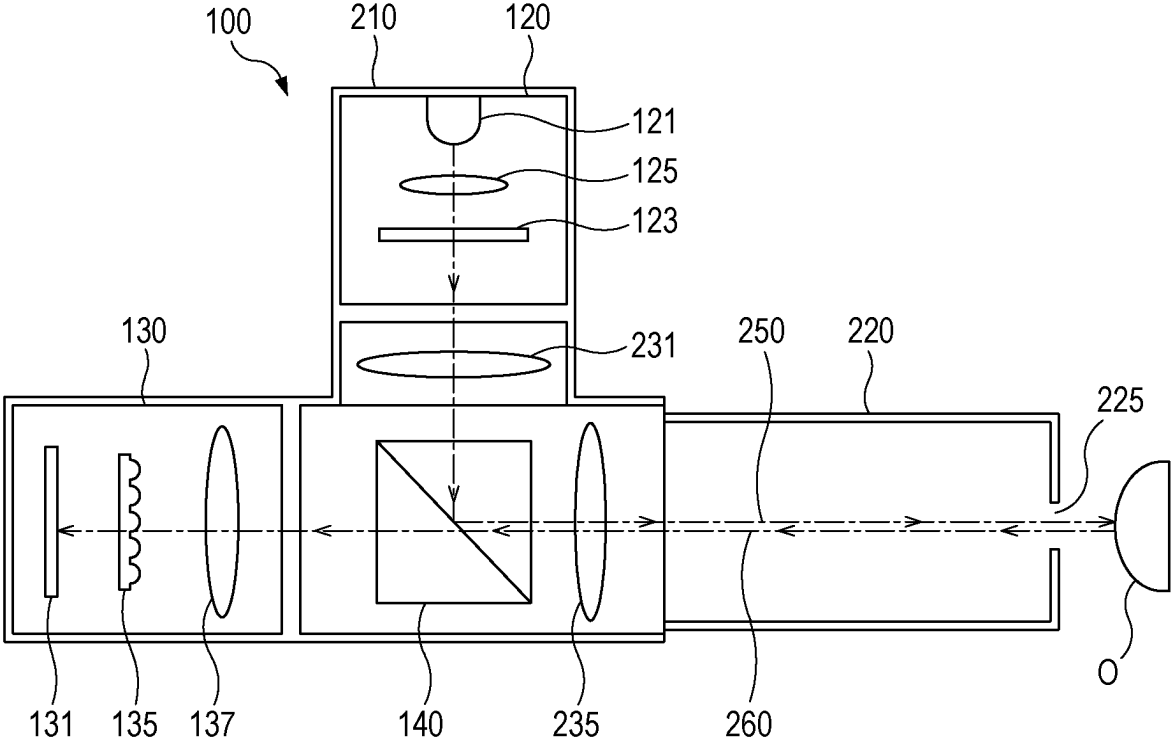


FIG. 3

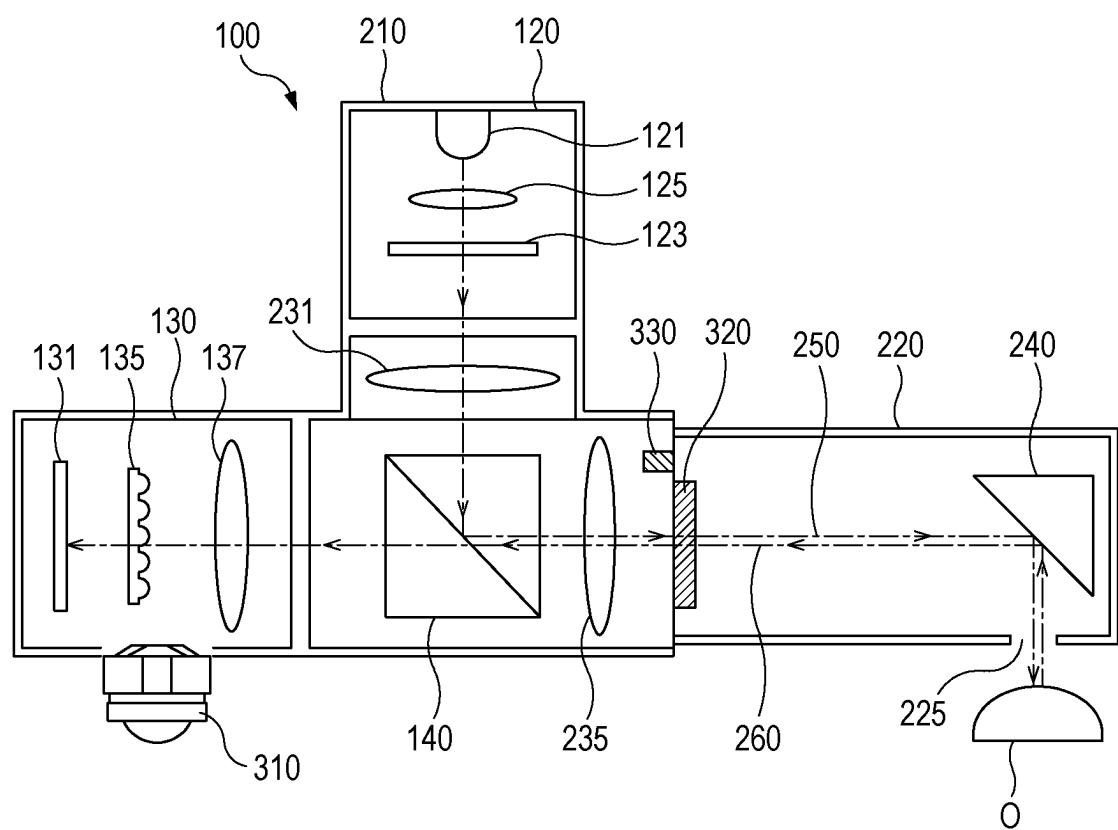


FIG. 4

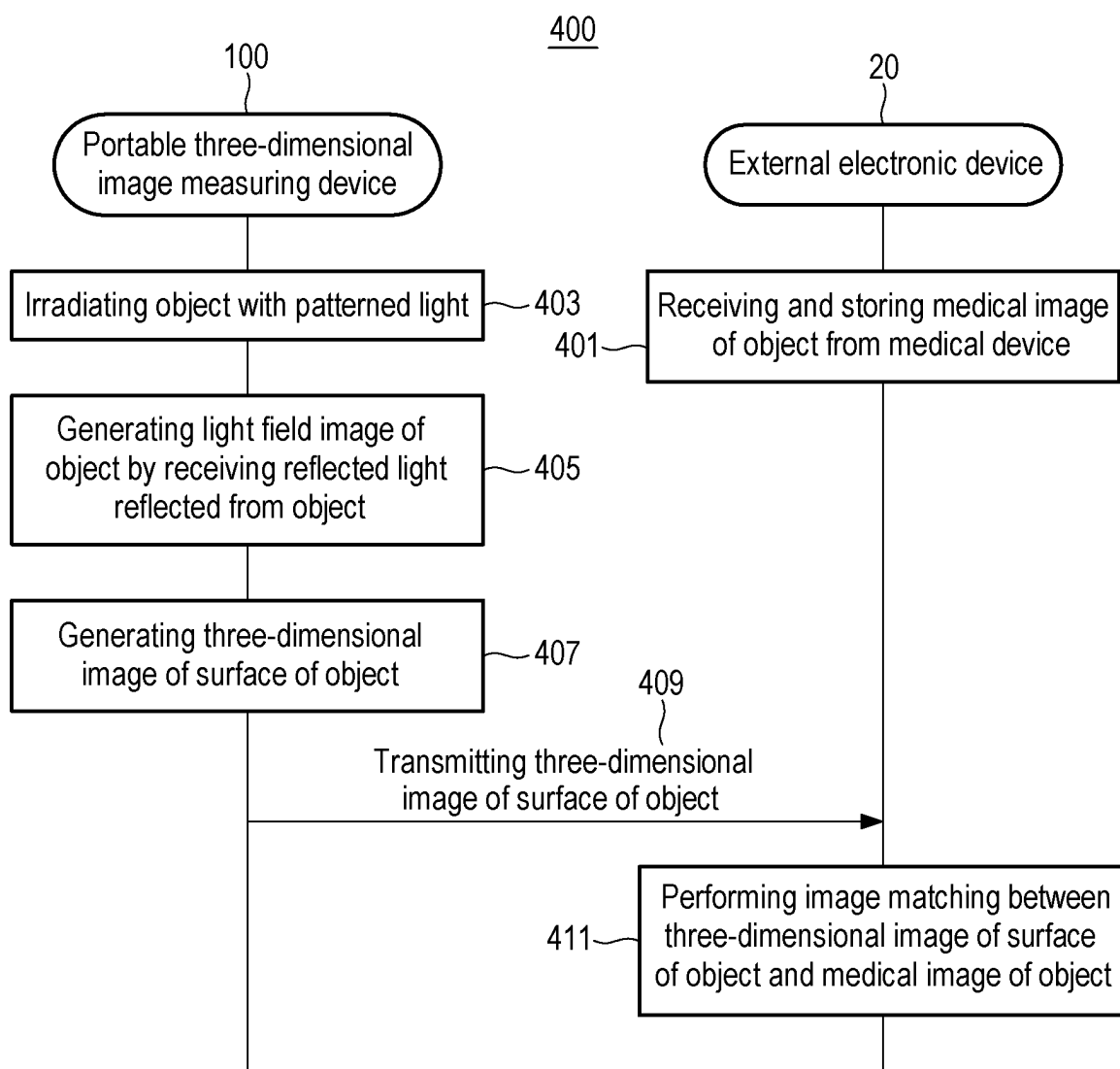


FIG. 5

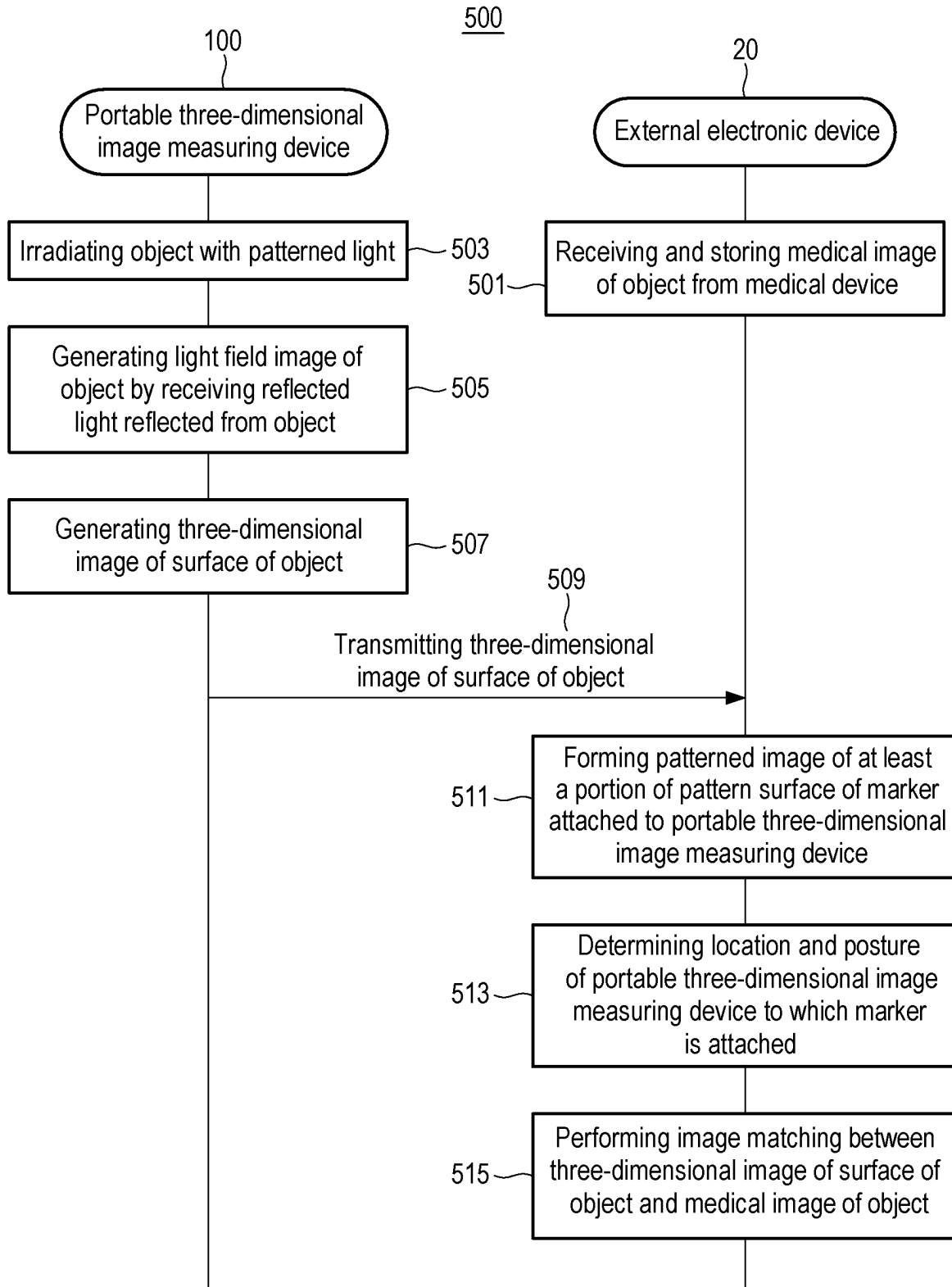


FIG. 6

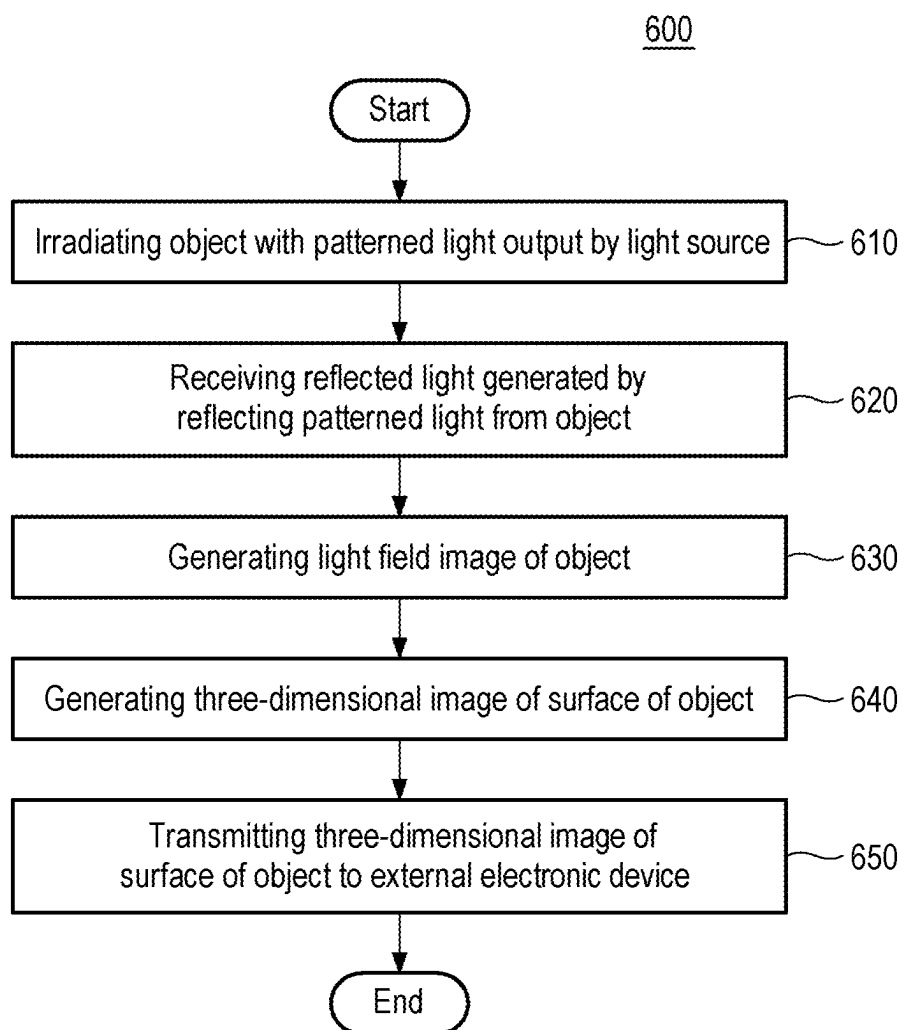


FIG. 7

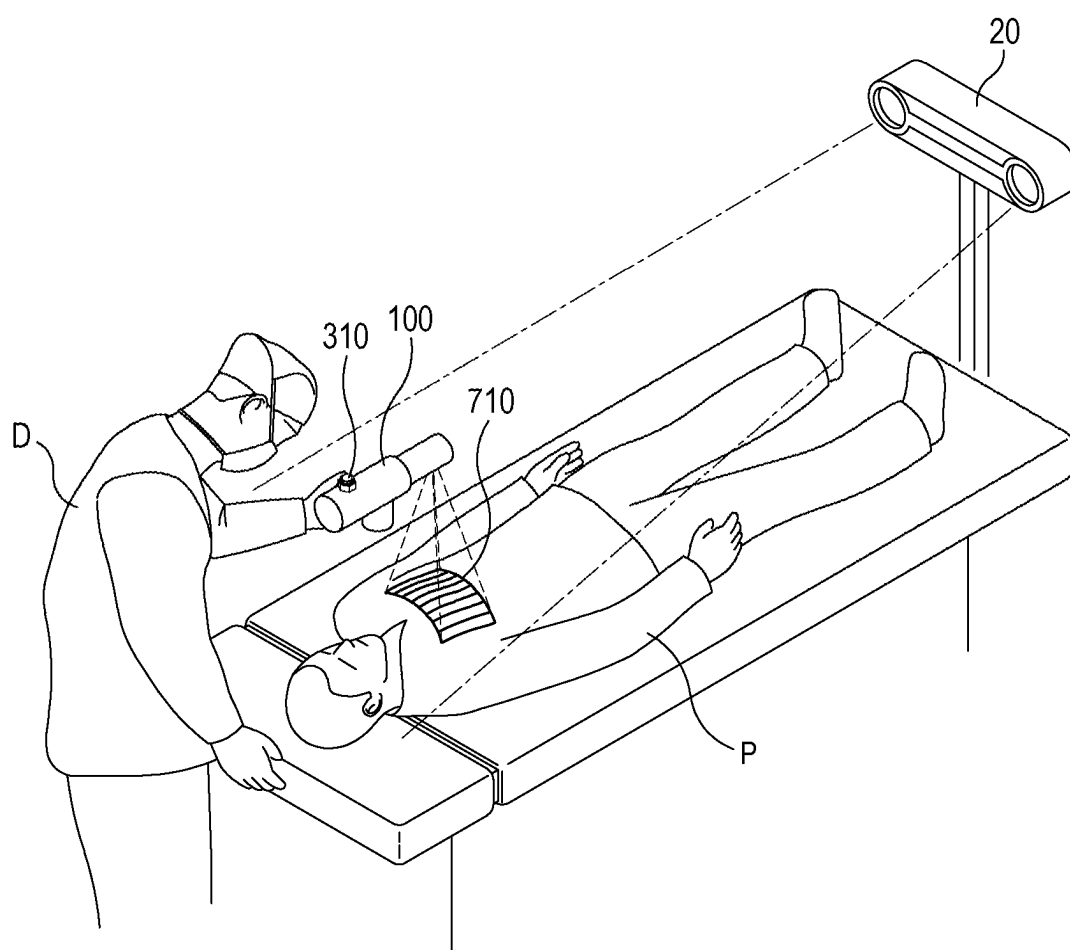


FIG. 8

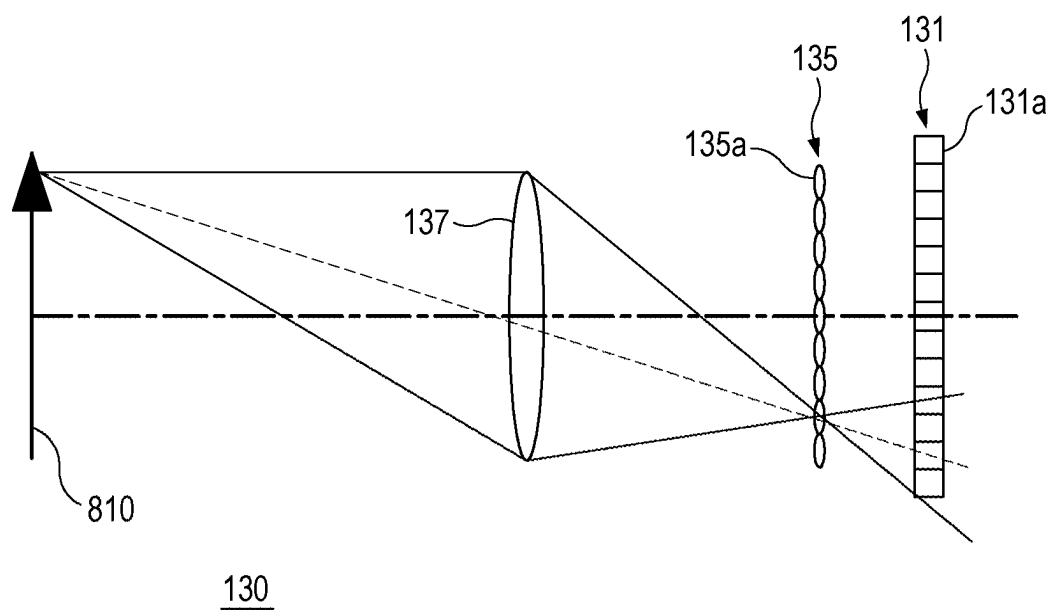


FIG. 9

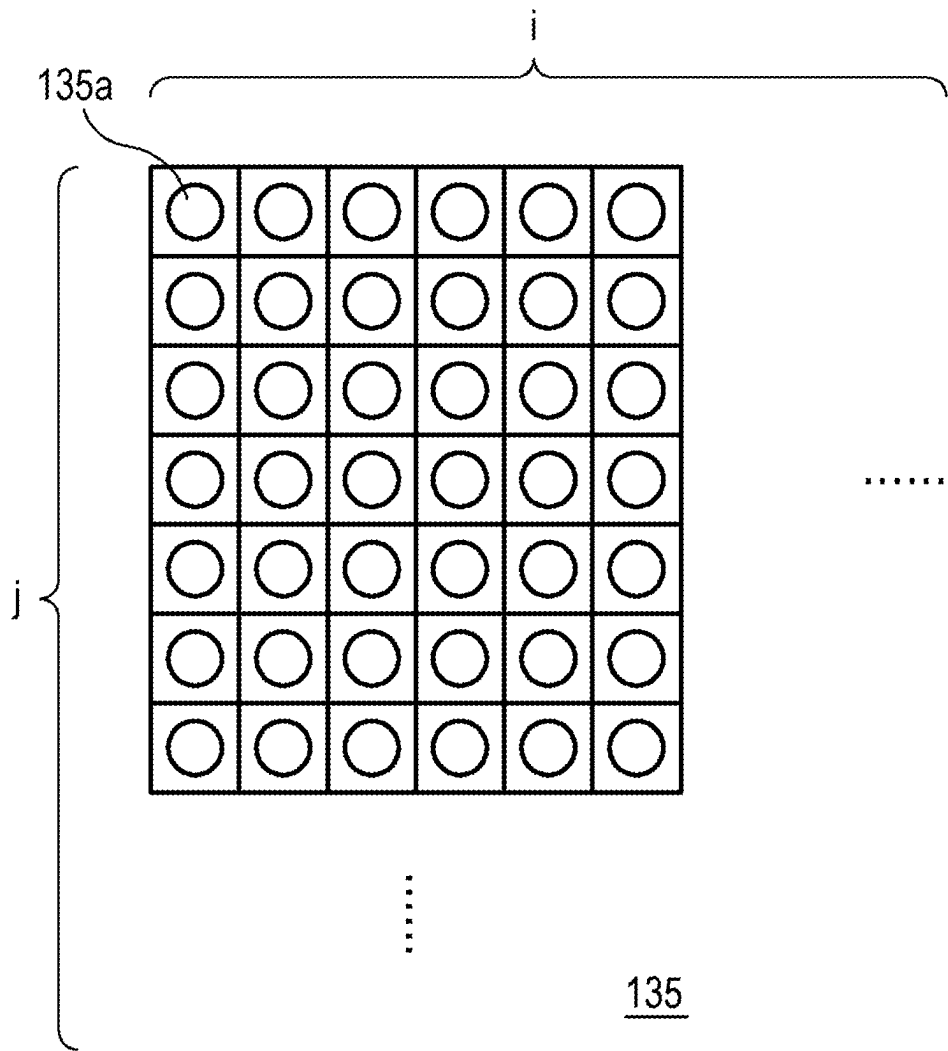


FIG. 10

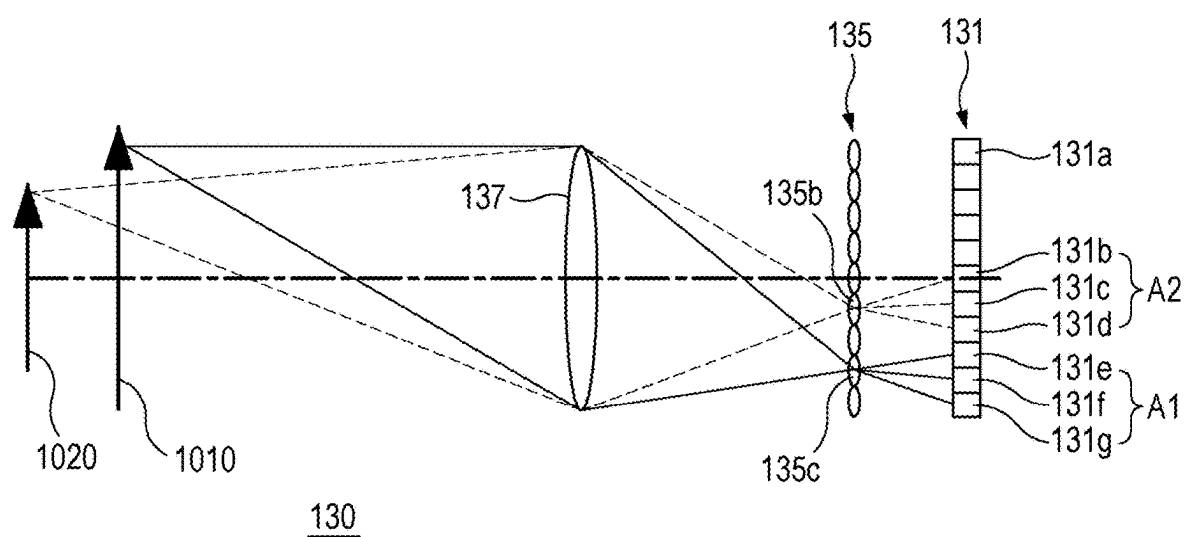
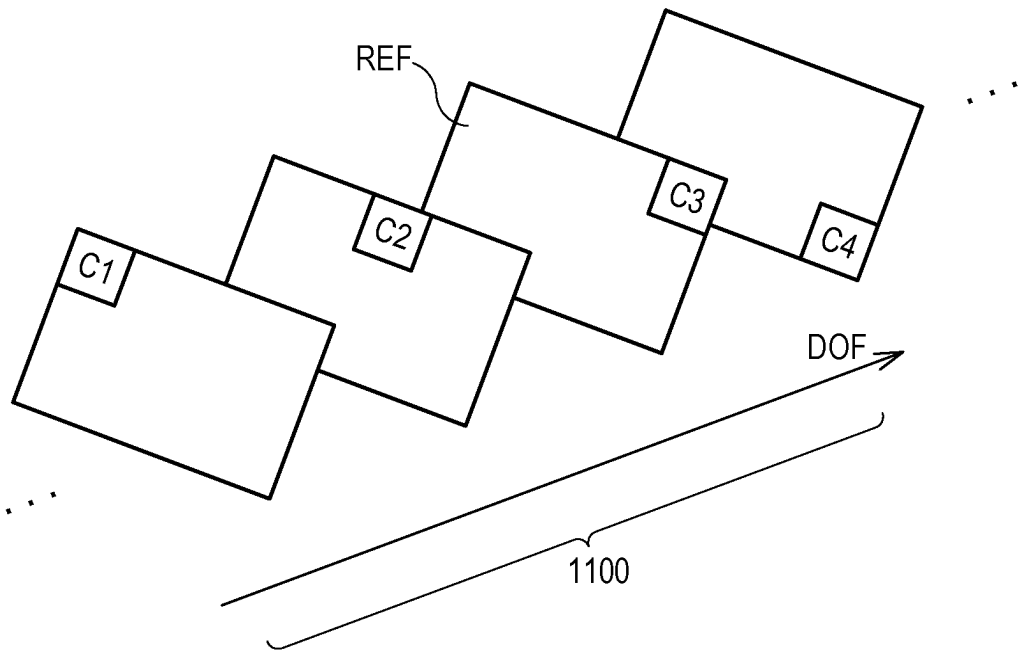


FIG. 11



**PORTABLE THREE-DIMENSIONAL IMAGE
MEASURING DEVICE,
THREE-DIMENSIONAL IMAGE
MEASURING METHOD USING SAME, AND
MEDICAL IMAGE MATCHING SYSTEM**

**CROSS-REFERENCE TO RELATED
APPLICATION(S)**

[0001] This is a continuation application of U.S. patent application Ser. No. 18/248,856 filed Apr. 12, 2023 (now pending), the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 18/248,856 is a national stage entry of International Application No. PCT/KR2021/014106, filed Oct. 13, 2021, which claims under 35 U.S.C. §§ 119(a) and 365(b) priority to and the benefits of Korean Patent Application No. 10-2020-0131843, filed Oct. 13, 2020, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a portable three-dimensional image measuring device. In particular, the present disclosure relates to a method of performing precise three-dimensional measurement of an object to be measured by providing a portable three-dimensional image measuring device. In addition, the present disclosure relates to a medical image matching system including the portable three-dimensional image measuring device.

[0003] This present disclosure is derived from research conducted as part of the WC300 project technology development support. [Project identification number: S2482672, Title of research project: Development of surgical navigation fusion head and neck surgery robot system with matching accuracy of 1 mm or less].

BACKGROUND

[0004] Various methods for measuring a three-dimensional image of an object are being used in the industry. Among them, a method of measuring a pattern generated by irradiating an object with certain patterned light, and obtaining a three-dimensional image of the object from the measured pattern is being used. For example, there is a moiré-type three-dimensional image measurement technology that measures a moiré pattern generated by irradiating an object with patterned light and obtains a three-dimensional image of the object from the moiré pattern.

[0005] Recently, surgical navigation technology has been used to support doctors in surgical operations. In general, the surgical navigation technology places markers on surgical tools and provides information in the form of displaying the location and posture information of the surgical tool on medical images (e.g., CT images and MRI images) of patients. Since the surgical navigation system needs to acquire and process three-dimensional image information on a specific affected part of a patient, a technique for measuring a three-dimensional image may be utilized.

SUMMARY

[0006] In the case of a three-dimensional image measuring device using a stereo camera, a three-dimensional image of an object may be measured through a triangulation method using a fixed pattern. The triangulation method is a method

of measuring a three-dimensional image of an object using two or more images captured at different locations. For example, the object may be irradiated with patterned light, images of the object irradiated with the patterned light may be acquired using two or more cameras disposed at different locations, and the three-dimensional image of the object may be acquired using the acquired images. The above three-dimensional image measuring device must include two or more cameras in order to use the triangulation method. The above three-dimensional image measuring device must include a light source that irradiates the object with the patterned light at different locations.

[0007] A three-dimensional image measuring device using a chromatic confocal sensor may measure a three-dimensional image of an object by measuring the depth of the object using chromatic aberration of a lens. The three-dimensional image measuring device using a chromatic confocal sensor must perform scanning of the side of the object to acquire the three-dimensional image of the object.

[0008] A portable three-dimensional image measuring device according to various embodiments of the present disclosure may include a light source configured to output patterned light, a camera configured to generate a light field image of an object by receiving reflected light generated by reflecting the patterned light from the object, and an optical path control element configured to reflect the patterned light output from the light source so that the object is irradiated with the patterned light, and to transmit the reflected light reflected from the object so that the reflected light reaches the camera. An optical path of the patterned light output from the light source and irradiated onto the object and an optical path of the reflected light reflected from the object and reaching the camera may overlap coaxially in a section between the optical path control element and the object.

[0009] A medical image matching system according to various embodiments of the present disclosure may include a portable three-dimensional image measuring device comprising a light source configured to output patterned light, a camera configured to generate a light field image of an object by receiving reflected light generated by reflecting the patterned light from the object, an optical path control element configured to reflect the patterned light output from the light source so that the object is irradiated with the patterned light, and to transmit the reflected light reflected from the object so that the reflected light reaches the camera, a communication circuit, and a processor, an optical path of the patterned light output from the light source and irradiated onto the object and an optical path of the reflected light reflected from the object and reaching the camera may overlap coaxially in a section between the optical path control element and the object, and the processor may be configured to generate a three-dimensional image of the surface of the object using the light field image of the object acquired through the camera and to transmit the three-dimensional image of the surface of the object to an external electronic device through the communication circuit.

[0010] A three-dimensional image measuring method of a portable three-dimensional image measuring device according to various embodiments of the present disclosure may include an operation of irradiating an object with patterned light output by a light source through an optical path control element and an operation of generating a light field image of the object by receiving reflected light, which is generated by reflecting the patterned light from the object, by a camera

through the optical path control element. An optical path of the patterned light output from the light source and irradiated onto the object and an optical path of the reflected light reflected from the object and reaching the camera may overlap coaxially in a section between the optical path control element and the object.

[0011] The three-dimensional image measuring device according to various embodiments of the present disclosure can generate a light field image of an object using one camera that generates a light field image and generate a three-dimensional image of the surface of the object by using the light field image of the object. Since the three-dimensional image measuring device according to various embodiments of the present disclosure can be implemented using only one camera, it can be miniaturized compared to a conventional stereo type three-dimensional image measuring device implemented to include two or more cameras. In addition, since it can be implemented using only one camera, the production cost of the three-dimensional image measuring device can be lowered, and the portability can be enhanced by reducing the weight.

[0012] According to various embodiments of the present disclosure, when photographing the object using the miniaturized three-dimensional image measuring device, it is easy for a user to move the three-dimensional image measuring device and to change the photographing posture of the three-dimensional image measuring device. In this case, the user can use the three-dimensional image measuring device to capture various postures (e.g., a lying down posture and a prone posture) of the object.

[0013] Since the three-dimensional image measuring device according to various embodiments of the present disclosure can use the camera that generates the light field image, the three-dimensional image of the object can be generated even with a single measurement.

[0014] In the three-dimensional image measuring device according to various embodiments of the present disclosure, since the optical axis of a light source emitting patterned light and the optical axis of a camera receiving light reflected from an object are coaxial in a partial section, distortion of an acquired light field image of the object can be minimized, and the device can be miniaturized. When the optical axis of the light source and the optical axis of the camera are coaxial in the partial section, the distortion of the light field image due to an inclination of the patterned light emitted from the light source may not occur.

[0015] In the three-dimensional image measuring device according to various embodiments of the present disclosure, since the optical axis of the light source emitting the patterned light and the optical axis of the camera receiving the light reflected from the object are coaxial in the partial section, the object can be uniformly irradiated with the patterned light, so that the loss of the amount of light can be minimized.

BRIEF DESCRIPTION OF DRAWINGS

[0016] FIG. 1 is a block diagram showing a medical image matching system according to various embodiments of the present disclosure.

[0017] FIGS. 2A and 2B are cross-sectional views of a portable three-dimensional image measuring device according to various embodiments.

[0018] FIG. 3 is a cross-sectional view of a portable three-dimensional image measuring device according to various embodiments.

[0019] FIG. 4 is an operation flowchart of the medical image matching system according to various embodiments of the present disclosure.

[0020] FIG. 5 is an operation flowchart of the medical image matching system according to various embodiments of the present disclosure.

[0021] FIG. 6 is an operation flowchart of the portable three-dimensional image measuring device according to various embodiments of the present disclosure.

[0022] FIG. 7 is a view showing an example in which the medical image matching system according to various embodiments of the present disclosure is used.

[0023] FIG. 8 is a view for explaining the structure of a camera according to various embodiments of the present disclosure.

[0024] FIG. 9 is a view for explaining a lens array of the camera according to various embodiments of the present disclosure.

[0025] FIG. 10 is a view illustrating a process of forming different subject depths of a plurality of sub-images included in a light field image acquired by the camera according to various embodiments of the present disclosure.

[0026] FIG. 11 is a view for explaining the light field image including the plurality of sub-images having different subject depths according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0027] Embodiments of the present disclosure are illustrated for describing the technical spirit of the present disclosure. The scope of the claims according to the present disclosure is not limited to the embodiments described below or to the detailed descriptions of these embodiments.

[0028] All technical or scientific terms used herein have meanings that are generally understood by a person having ordinary knowledge in the art to which the present disclosure pertains, unless otherwise specified. The terms used herein are selected for only more clear illustration of the present disclosure, and are not intended to limit the scope of claims in accordance with the present disclosure.

[0029] The expressions “include,” “provided with,” “have” and the like used herein should be understood as open-ended terms connoting the possibility of inclusion of other embodiments, unless otherwise mentioned in a phrase or sentence including the expressions.

[0030] A singular expression can include meanings of plurality, unless otherwise mentioned, and the same is applied to a singular expression recited in the claims.

[0031] The terms “first,” “second,” etc. used herein are used to identify a plurality of components from one another, and are not intended to limit the order or importance of the relevant components.

[0032] The term “unit” used in these embodiments means a software component or hardware component, such as a field-programmable gate array (FPGA) and an application specific integrated circuit (ASIC). However, a “unit” is not limited to software and hardware, it may be configured to be an addressable storage medium or may be configured to run on one or more processors 110. For example, a “unit” may include components, such as software components, object-oriented software components, class components, and task

components, as well as processors **110**, functions, attributes, procedures, subroutines, segments of program codes, drivers, firmware, micro-codes, circuits, data, databases, data structures, tables, arrays, and variables. Functions provided in components and “unit” may be combined into a smaller number of components and “units” or further subdivided into additional components and “units.”

[0033] The expression “based on” used herein is used to describe one or more factors that influence a decision, an action of judgment or an operation described in a phrase or sentence including the relevant expression, and this expression does not exclude additional factor influencing the decision, the action of judgment or the operation.

[0034] When a certain component is described as being “coupled to” or “connected to” another component, this should be understood as meaning that the certain component may be coupled or connected directly to the other component or that the certain component may be coupled or connected to the other component via a new intervening component.

[0035] Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. In the accompanying drawings, like or relevant components are indicated by like reference numerals. In the following description of embodiments, repeated descriptions of the identical or relevant components will be omitted. However, even if a description of a component is omitted, such a component is not intended to be excluded in an embodiment.

[0036] Although process steps, method steps, algorithms, etc. are described in a sequential order in the shown flowcharts, such processes, methods, and algorithms may be configured to operate in any suitable order. In other words, the steps of the processes, methods, and algorithms described in various embodiments of the present disclosure do not have to be performed in the order described in the present disclosure. In addition, although some steps are described as being performed asynchronously, they may be performed concurrently in other embodiments. Further, illustration of processes by depiction in the drawings does not mean that the illustrated processes are exclusive of other changes and modifications thereto, that any of the illustrated processes or steps thereof are essential to one or more of various embodiments of the present disclosure, and that the illustrated processes are preferable.

[0037] FIG. 1 is a block diagram showing a medical image matching system **10** according to various embodiments of the present disclosure.

[0038] Referring to FIG. 1, the medical image matching system **10** according to various embodiments may include a portable three-dimensional image measuring device **100** and an external electronic device **20**. The portable three-dimensional image measuring device **100** and the external electronic device **20** may communicate with each other to transmit/receive various data (e.g., images). Even if some of the components shown in FIG. 1 are omitted or replaced, there will be no problem in implementing various embodiments disclosed in the present disclosure.

[0039] The portable three-dimensional image measuring device **100** according to various embodiments may include a processor **110**, a light source **120**, a camera **130**, an optical path control element **140**, and a communication circuit **150**.

The portable three-dimensional image measuring device **100** may further include a first housing (not shown) and a second housing (not shown).

[0040] The processor **110** according to various embodiments may be a component capable of performing calculations or data processing related to control and/or communication of other components of the portable three-dimensional image measuring device **100**. The processor **110** may be operatively connected to other components of the portable three-dimensional image measuring device **100**, for example. The processor **110** may load commands or data received from other components of the portable three-dimensional image measuring device **100** into a memory (not shown), process the commands or data stored in the memory, and output the resulting data.

[0041] The light source **120** according to various embodiments may output patterned light. The light source **120** may irradiate an object with the patterned light. The patterned light may be light having a specific pattern or light having a pattern with a constant or specific period in order to measure a three-dimensional image of the object. The patterned light may include, for example, patterned light in the form of random dots, patterned light in the form of a checkered pattern, patterned light in which the brightness of stripes is in the form of a sine wave, patterned light in the form of on-off with repeated bright and dark portions, and triangular wave patterned light in which a change in brightness is a triangular waveform. However, this is only for the purpose of explanation, and the shape of the patterned light is not limited thereto.

[0042] The light source **120** according to various embodiments may include a pattern unit in which a plurality of patterns are formed, and an LED that irradiates the pattern unit with light. The light source **120** may further include a condensing lens configured to condense the light output from the LED and irradiate the pattern unit with the condensed light. The light output from the LED may reflect the patterns by passing through the pattern unit in which the plurality of patterns are formed. The LED may emit infrared light, for example, but is not limited thereto.

[0043] The camera **130** according to various embodiments may be a component that captures an image of an object. The camera **130** may acquire image data of the object by photographing the object, and may acquire a three-dimensional image of the object by processing the acquired image data. For example, the camera **130** may acquire the image of the object by photographing the object irradiated with the patterned light. The processor **110** may generate the three-dimensional image of the object based on a phase shift method using the patterned light. For example, when an object is irradiated with the patterned light of a certain shape through the light source **120**, the intensity of light appearing on the surface of the object may vary according to the curvature of the surface of the object. In this case, the camera **130** may generate a light field image of the object reflected with the pattern, and the processor **110** may generate a three-dimensional image for the surface of the object by generating phase data from the light field image and calculating the height of each point constituting the surface of the object.

[0044] The camera **130** according to various embodiments may be a light field camera **130** that generates a light field image. The light field camera **130** may be configured to determine the depth of the object posteriorly after photo-

graphing the object, and to combine images having different depths of the object. An image sensor of the light field camera 130 may have a posterior and variable object depth.

[0045] The camera 130 according to various embodiments may include a condensing lens, a lens array, and an image sensor. The condensing lens may, for example, condense light entering from an object. The lens array may be, for example, a lens in which a plurality of micro lenses are arranged. The image sensor may, for example, capture light passing through the lens array and generate a light field image using the captured light. The image sensor may be divided into regions corresponding to the respective ones of the plurality of micro lenses. The image sensor may include, for example, a charge-coupled device (CCD) sensor or a complementary metal-oxide semiconductor (CMOS) sensor. A detailed description of each component included in the camera 130 will be described in FIGS. 8 to 11.

[0046] The light field image generated by the camera 130 according to various embodiments may include a plurality of sub-images that store color information and direction information of light together. For example, when an object is irradiated with patterned light and reflected light reflected from the object is received by the camera 130, the light field image may be an image in which a plurality of sub-images including color information and direction information of the reflected light are combined. The camera 130 may perform a refocusing process using the plurality of sub-images included in the light field image. For example, in the refocusing process, the camera 130 may generate an image of a desired depth by combining the depth of a desired object and color information of pixels corresponding to the optical path and direction backward calculated accordingly, among pixels of the light field image. For example, the camera 130 may generate an image in which all regions of the object are in focus during the refocusing process. In order for the camera 130 to form an image of an accurate photographing target region, a distance between the portable three-dimensional image measuring device 100 and the photographing target region of the object needs to be appropriately adjusted. When using the camera 130 that generates the light field image, since the depth of the object can be determined posteriorly and a focused light field image can be generated for all regions of the object, there is no need to adjust the focal length in advance. In the case of the camera 130 generating the light field image, a measurable depth range is wider than that of the camera 130 using a general lens, so that a three-dimensional image of the object can be acquired with one shot.

[0047] The optical path control element 140 according to various embodiments may reflect the patterned light in a specific direction so that the object is irradiated with the patterned light output from the light source 120. The optical path control element 140 may transmit the reflected light reflected from the object to reach the camera 130. The optical path control element 140 may be, for example, a transmissive mirror. According to various embodiments, the light source 120 and the camera 130 may be disposed perpendicular to each other with respect to the optical path control element 140.

[0048] The light source 120, the camera 130, and the optical path control element 140 may be disposed inside the first housing according to various embodiments. The second housing according to various embodiments may be coupled to the first housing and may have an opening through which

the object is irradiated with the patterned light output from the light source 120. The second housing may be rotatably coupled to the first housing. The first housing or the second housing may be coupled with a component (e.g., a handle) that facilitates a user's movement, carrying, and use of the portable three-dimensional image measuring device 100.

[0049] According to various embodiments, the communication circuit 150 may establish a communication channel with the external electronic device 20 and transmit/receive various data to/from the external electronic device 20. According to various embodiments, the communication circuit 150 may include a cellular communication module and be configured to be connected to a cellular network (e.g., 3G, LTE, 5G, Wibro, or Wimax). According to various embodiments, the communication circuit 150 may include a short-range communication module to transmit/receive data to/from the external electronic device 20 by using short-range communication (e.g., Wi-Fi, Bluetooth, Bluetooth Low Energy (BLE), or UWB), but it is not limited thereto.

[0050] The processor 110 according to various embodiments may generate a three-dimensional image of the surface of the object using the light field image of the object acquired through the camera 130. For example, the light intensity of the emitted patterned light displayed on the surface of the target region to be actually photographed may vary according to the curvature of the surface of the target region of the object to be photographed. The processor 110 may use the light field image of the object to measure the intensity of light that varies according to the curvature of the surface of the object, generate phase data from the measured light intensity, and calculate the height of each point constituting the surface. The processor 110 may generate a three-dimensional image of the surface of the object by calculating the height of each point constituting the surface of the object.

[0051] The processor 110 according to various embodiments may transmit the three-dimensional image of the surface of the object to the external electronic device 20 through the communication circuit 150.

[0052] The external electronic device 20 according to various embodiments may include a controller 21, an imaging device 23, a storage 25, and a communication circuit 27. The controller 21 according to various embodiments may be a component capable of performing calculations or data processing related to control and/or communication of other components of the external electronic device 20. The controller 21 may be operatively connected to other components of the external electronic device 20, for example.

[0053] The imaging device 23 according to various embodiments may image at least a portion of a pattern surface of a marker (not shown) attached to the portable three-dimensional image measuring device 100 to form a pattern image of the at least a portion of the pattern surface. The imaging device 23 may include, for example, at least two or more cameras capable of forming an image of at least a portion of the marker. The external electronic device 20 may determine the location and/or posture of the marker or the portable three-dimensional image measuring device 100 to which the marker is attached, using the formed pattern image.

[0054] For example, when the pattern image of the marker is acquired, the external electronic device 20 may extract at least one among sub-patterns, from the pattern image as a basic unit constituting the marker pattern. The location of at

least one extracted sub-pattern within the entire pattern may be determined, and the posture of the marker may be determined based on the determined location of the sub-pattern within the entire pattern. Here, the posture of the marker may mean a three-dimensional direction or orientation of the marker relative to the imaging device **23**. For example, the location of the marker or the portable three-dimensional image measuring device **100** may be determined using triangulation based on two images having a stereoscopic relationship among images formed by the imaging device **23** including at least two cameras. When the location and posture of the marker are determined as described above, the location and posture of the portable three-dimensional image measuring device **100** to which the marker is attached may be determined based on the geometric relationship between the marker and the portable three-dimensional image measuring device **100** to which the marker is attached.

[0055] The storage **25** according to various embodiments may store various data used by at least one component (e.g., the controller **21**) of the external electronic device **20**. For example, the controller **21** may cause the storage **25** to store the three-dimensional image of the surface of the object received from the portable three-dimensional image measuring device **100**. For example, the controller **21** may cause the storage **25** to store medical images (e.g., CT images and MRI images) received from a medical device (not shown).

[0056] The communication circuit **27** of the external electronic device **20** according to various embodiments may establish a communication channel with the portable three-dimensional image measuring device **100** and transmit/receive various data to/from the portable three-dimensional image measuring device **100**. According to various embodiments, the communication circuit **27** of the external electronic device **20** may include a cellular communication module and be configured to be connected to a cellular network (e.g., 3G, LTE, 5G, Wibro, or Wimax). According to various embodiments, the communication circuit **27** of the external electronic device **20** includes a short-range communication module to transmit/receive data to/from the portable three-dimensional image measuring device **100** by using short-range communication (e.g., Wi-Fi, Bluetooth, Bluetooth Low Energy (BLE), or UWB), but it is not limited thereto.

[0057] The controller **21** of the external electronic device **20** according to various embodiments may perform image matching between the three-dimensional image of the surface of the object received from the portable three-dimensional image measuring device **100** and the medical image of the object. The three-dimensional image of the surface of the object generated by the portable three-dimensional image measuring device **100** may be the external surface of a target included in the medical image or a portion thereof. For example, if the medical image is an image modeling the three-dimensional shape of a head of the object, the three-dimensional image of the surface of the object may be an image obtained by measuring the external shapes of the eyes, nose, mouth, ears, etc. on the surface of the object's head.

[0058] According to various embodiments, the three-dimensional image of the surface of the object may have a unique coordinate system (for example, x1y1z1 coordinate system) of the portable three-dimensional image measuring device **100**. The coordinate system of the three-dimensional

image of the surface of the object may be different from the coordinate system of the medical image (for example, x2y2z2 system) and the coordinate system of the external electronic device **20** (for example, x0y0z0 system). The coordinate system of the external electronic device **20** may mean, for example, the coordinate system of the imaging device of the external electronic device **20**. A detailed image matching method will be described with reference to FIG. **4** later.

[0059] FIGS. **2A** and **2B** are cross-sectional views of the portable three-dimensional image measuring device **100** according to various embodiments. Specifically, FIGS. **2A** and **2B** are views schematically showing only some of the components of the portable three-dimensional image measuring device **100** in order to explain the arrangement relationship between the components. Contents overlapping with those described in FIG. **1** will be omitted.

[0060] Referring to FIG. **2A**, the portable three-dimensional image measuring device **100** according to various embodiments may include the light source **120**, the camera **130**, and the optical path control element **140**. The portable three-dimensional image measuring device **100** may include a first housing **210** in which the light source **120**, the camera **130**, and the optical path control element **140** are disposed, and a second housing **220** that is coupled to the first housing **210** and is formed with an opening **225** through which an object **O** is irradiated with the patterned light output from the light source **120**. The light source **120** and the camera **130** according to various embodiments may be disposed perpendicular to each other with respect to the optical path control element **140**. According to various embodiments, at least one condensing lens **231** or **235** for condensing light may be disposed around the optical path control element **140**.

[0061] The light source **120** according to various embodiments may include a pattern unit **123** in which a plurality of patterns are formed, and an LED **121** that irradiates the pattern unit **123** with light. The light source **120** may further include a condensing lens **125**, which is configured to condense the light output from the LED **121** and irradiate the pattern unit **123** with the condensed light, between the pattern unit **123** and the LED **121**. The light output from the LED **121** may reflect the patterns by passing through the pattern unit **123**. According to various embodiments, the patterned light output from the light source **120** may be incident to the optical path control element **140**. The patterned light incident to the optical path control element **140** may be reflected toward the second housing **220** so that the object **O** can be irradiated with the patterned light. The patterned light incident into the second housing **220** may be reflected by a reflective mirror **240** so that the object **O** can be irradiated with the patterned light through the opening **225** of the second housing **220**.

[0062] According to various embodiments, the patterned light with which the object **O** is irradiated may be reflected by the object **O**. The reflected light reflected from the object **O** may be incident again into the second housing **220** through the opening **225**. The reflected light may be reflected by the reflective mirror **240** and incident to the optical path control element **140**. The reflected light incident to the optical path control element **140** may pass through the optical path control element **140** and reach the camera **130**. The reflected light passing through the optical path control element **140** may pass through a condensing lens **137** and pass through a lens array **135** in which a plurality of micro

lenses are arranged. An image sensor **131** may capture the reflected light passing through the lens array **135**. The image sensor **131** may generate a light field image of the object **O** by capturing the reflected light. The light field image of the object **O** may be an image of patterned light with which the object **O** is irradiated. The processor **110** may generate a three-dimensional image of the surface of the object **O** by using the light field image of the object **O**, and may transmit the three-dimensional image of the surface of the object **O** to the external electronic device **20** through the communication circuit **150**.

[0063] According to various embodiments, an optical path **250** of the patterned light which is output from the light source **120** and irradiated onto the object **O** and an optical path **260** of the reflected light which is reflected from the object **O** and reaches the camera **130** may overlap in a coaxial manner in a section between the optical path control element **140** and the object **O**. When the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of the reflected light reflected from the object **O** are coaxial, the portable three-dimensional image measuring device **100** can be miniaturized and an accurate image of the object **O** may be acquired. For example, if the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of the reflected light reflected from the object **O** are not coaxial, the patterned light may obliquely irradiate the object **O** or the reflected light reflected from the object **O** may obliquely reach the camera **130**. A pattern formed on the object **O** by the patterned light irradiate obliquely may have a distorted shape compared to a pattern formed on the object **O** by the patterned light irradiate perpendicularly. In this case, a distorted image of the object **O** may be acquired. On the other hand, when the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of the reflected light reflected from the object **O** are coaxial, a user may use the three-dimensional image measuring device **100** to acquire an accurate and undistorted image of the object **O**.

[0064] According to various embodiments, when the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of the reflected light reflected from the object **O** are coaxial, the portable three-dimensional image measuring device **100** may easily photograph the object **O**. When photographing the object **O** using the miniaturized three-dimensional image measuring device **100**, the user can easily move the three-dimensional image measuring device **100** and easily change the photographing posture of the three-dimensional image measuring device **100**. In the above case, the user may use the three-dimensional image measuring device **100** to capture various postures (e.g., a lying down posture and a prone posture) of the object.

[0065] According to various embodiments, when the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of the reflected light reflected from the object **O** are coaxial, the portable three-dimensional image measuring device **100** may acquire the image of the object using one camera **130** instead of acquiring the image of the object through the triangulation method using two or more cameras. Therefore, the portable three-dimensional image measuring device **100** can be miniaturized compared to the conventional stereo-type 3D image measuring device implemented by including two or more

cameras, which may result in reduction of production cost and enhanced portability by reducing weight.

[0066] In the three-dimensional image measuring device **100** according to various embodiments, since the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of the reflected light reflected from the object **O** are coaxial, the object **O** may be uniformly irradiated with the patterned light, and loss of light quantity may be minimized.

[0067] Referring to FIG. 2B, the portable three-dimensional image measuring device **100** according to various embodiments may not include the separate reflective mirror **240** shown in FIG. 2A. In this case, the object **O** may be irradiated with the patterned light reflected from the optical path control element **140** through the opening **225** formed on the optical path **250** of the patterned light without additional reflection. In addition to the structure shown in FIGS. 2A and 2B, a variety of structures in which the optical path **250** of the patterned light with which the object **O** is irradiated and the optical path **260** of reflected light reflected from the object **O** are coaxial may be applied to the portable three-dimensional image measuring device **100**.

[0068] FIG. 3 is a cross-sectional view of the portable three-dimensional image measuring device **100** according to various embodiments. Specifically, FIG. 3 is a view schematically showing only some of the components of the portable three-dimensional image measuring device **100** in order to explain the arrangement relationship between the components. Contents overlapping with those described in FIG. 2 are omitted.

[0069] The portable three-dimensional image measuring device **100** according to various embodiments may further include a marker **310**. For example, the marker **310** may be attached to the first housing **210** of the portable three-dimensional image measuring device **100**. The marker **310** may include a pattern surface on which a pattern is formed and a lens configured to identify at least a portion of the pattern uniquely appearing from the outside of the marker **310** according to a direction viewed from the outside of the marker **310**. The lens of the marker **310** may be a ball lens, and the pattern surface may have a curved shape.

[0070] The external electronic device **20** according to various embodiments may image at least a portion of the pattern surface of the marker **310** through the imaging device **23** to form a pattern image for at least a portion of the pattern surface. The external electronic device **20** may determine the location or coordinate and posture or orientation of the portable three-dimensional image measuring device **100** to which the marker **310** is attached, based on the formed pattern image.

[0071] The location of the portable three-dimensional image measuring device **100** may be defined by spatial coordinates such as coordinates on the x, y, and z axes of the Cartesian coordinate system. The posture of the portable three-dimensional image measuring device **100** may be defined as roll, pitch, and yaw. The external electronic device **20** may capture an image of the marker **310** attached to the portable three-dimensional image measuring device **100** through the imaging device **23**, thereby tracking the location and posture of the portable three-dimensional image measuring device **100**.

[0072] For example, the external electronic device **20** may form an image of at least a portion of the pattern surface of the marker **310** attached to the portable three-dimensional

image measuring device **100** through the imaging device **23**. For example, the imaging device **23** of the external electronic device **20** may form a pattern image of at least a portion of a pattern visually identified from the outside of the marker **310** through a ball lens of the marker **310**. When the pattern image of at least a portion of the pattern surface is acquired, the external electronic device **20** may process information extracted from the pattern image of at least the portion of the pattern surface to determine the location and posture of the marker **310**. The external electronic device **20** may determine the location and posture of the portable three-dimensional image measuring device **100** to which the marker **310** is attached, based on the location and posture of the marker **310**. A specific method of calculating the location and posture of the marker **310** using the image of at least the portion of the pattern surface may be the same as a general optical tracking method.

[0073] The marker **310** according to various embodiments may be installed to be movable from a predetermined location of the portable three-dimensional image measuring device **100**. The processor **110** may transmit information, which indicates the displacement of the marker **310** from the predetermined location of the portable three-dimensional image measuring device **100**, to the external electronic device **20** through the communication circuit **150**. The external electronic device **20** receives the information indicating the displacement of the marker **310** and may correct the location or posture of the portable three-dimensional image measuring device **100** based on the received information indicating the displacement of the marker **310**. Information on the corrected location or posture of the portable three-dimensional image measuring device **100** may be used for image matching between the three-dimensional image of the surface of the object **O** and a medical image.

[0074] The three-dimensional image of the surface of the object **O** may have a unique coordinate system (for example, $x_1y_1z_1$ coordinate system) of the portable three-dimensional image measuring device **100**. The coordinate system of the three-dimensional image of the surface of the object **O** may be different from the coordinate system (for example, $x_2y_2z_2$) of the medical image and may be different from the coordinate system (for example, $x_0y_0z_0$) of the external electronic device **20**.

[0075] The medical image matching system **10** according to various embodiments may transform or align the coordinate system (for example, $x_2y_2z_2$) of the medical image and the coordinate system (for example, $x_1y_1z_1$) of the three-dimensional image of the surface of the object **O** into the coordinate system (for example, $x_0y_0z_0$) of the external electronic device **20**. The external electronic device **20** may perform matching between the medical image and the three-dimensional image of the surface of the object **O**, which have different coordinate systems. In order to match between the medical image and the three-dimensional image of the surface of the object **O**, the external electronic device **20** may extract a surface image from the medical image and perform matching between the extracted surface image and the received three-dimensional image of the surface of the object **O**. Here, the surface image extracted from the medical image may have the same coordinate system (for example, $x_2y_2z_2$) of the medical image. In addition, the external electronic device **20** may transform the coordinate system (for example, $x_1y_1z_1$) of the three-dimensional image of the

surface of the object **O** into the coordinate system (for example, $x_0y_0z_0$) of the external electronic device **20** by mediation of the marker **310** attached to the portable three-dimensional image measuring device **100**. In addition, the medical image and the surface image extracted from the medical image may be transformed into the coordinate system (for example, $x_0y_0z_0$) of the external electronic device **20**. The external electronic device **20** may perform matching between the three-dimensional image of the surface of the object **O** and the medical image using various image matching algorithms. For example, the external electronic device **20** may perform matching using an interactive closest point (ICP) algorithm.

[0076] The portable three-dimensional image measuring device **100** according to various embodiments may further include a bearing **320** and a sensor **330**. The second housing **220** according to various embodiments may be rotatably coupled to the first housing **210**. The bearing **320** may be a mechanical element that rotatably couples the second housing **220** to the first housing **210**. The second housing **220** may rotate based on the central axis of the bearing **320** and may rotate independently of the first housing **210**. The sensor **330** may be a sensor that senses an angle at which the second housing **220** rotates with respect to the first housing **210**. The sensor **330** may be, for example, a gyro sensor or an encoder. The processor **110** according to various embodiments may transmit information on the angle at which the second housing **220** rotates with respect to the first housing **210**, to the external electronic device **20** through the communication circuit **150**.

[0077] FIG. **4** is an operation flowchart of the medical image matching system **10** according to various embodiments of the present disclosure.

[0078] Referring to the operation flowchart **400**, in operation **401**, the external electronic device **20** according to various embodiments may receive and store a medical image of an object from a medical device. The medical image may be, for example, a CT image or an MRI image.

[0079] In operation **403**, the portable three-dimensional image measuring device **100** according to various embodiments may irradiate the object with patterned light. For example, the portable three-dimensional image measuring device **100** may output patterned light through the light source **120**. The patterned light output through the light source **120** may be reflected by the optical path control element **140** and the object may be irradiated with the patterned light.

[0080] In operation **405**, the portable three-dimensional image measuring device **100** according to various embodiments may generate a light field image of the object by receiving reflected light reflected from the object. For example, the reflected light reflected from the object may reach the camera **130** through the optical path control element **140**. The camera **130** may generate the light field image of the object by receiving the reflected light.

[0081] In operation **407**, the portable three-dimensional image measuring device **100** according to various embodiments may generate a three-dimensional image of the surface of the object by using the light field image of the object. For example, the processor **110** may measure the light intensity of patterned light included in the light field image of the object and generate phase data based on the measured light intensity of the patterned light. The processor **110** may generate the three-dimensional image of the surface of the

object by calculating the height of each point constituting the surface of the object based on the generated phase data.

[0082] In operation 409, the portable three-dimensional image measuring device 100 according to various embodiments may transmit the three-dimensional image of the surface of the object to the external electronic device 20. For example, the processor 110 of the portable three-dimensional image measuring device 100 may transmit the three-dimensional image of the surface of the object to the external electronic device 20 through the communication circuit 150.

[0083] In operation 411, the external electronic device 20 according to various embodiments may perform image matching between the three-dimensional image of the surface of the object received from the portable three-dimensional measurement device and a pre-stored medical image of the object.

[0084] FIG. 5 is an operation flowchart of the medical image matching system according to various embodiments of the present disclosure. Contents overlapping with those described in FIG. 4 are omitted.

[0085] Referring to the operation flowchart 500, in operation 501, the external electronic device 20 according to various embodiments may receive and store a medical image of an object from a medical device.

[0086] In operation 503, the portable three-dimensional image measuring device 100 according to various embodiments may irradiate the object with patterned light.

[0087] In operation 505, the portable three-dimensional image measuring device 100 according to various embodiments may generate a light field image of the object by receiving reflected light reflected from the object.

[0088] In operation 507, the portable three-dimensional image measuring device 100 according to various embodiments may generate a three-dimensional image of the surface of the object by using the light field image of the object.

[0089] In operation 509, the portable three-dimensional image measuring device 100 according to various embodiments may transmit the three-dimensional image of the surface of the object to the external electronic device 20.

[0090] In operation 511, the external electronic device 20 according to various embodiments may form an image of at least a portion of the pattern surface of the marker 310 attached to the portable three-dimensional image measuring device 100. For example, the external electronic device 20 may form a pattern image of at least the portion of the pattern surface of the marker 310 through an imaging device.

[0091] In operation 513, the external electronic device 20 according to various embodiments may determine the location and posture of the portable three-dimensional image measuring device 100 to which the marker 310 is attached, based on the formed pattern image. For example, the external electronic device 20 may process information extracted from the pattern image of at least the portion of the pattern surface to determine the location and posture of the marker 310. The external electronic device 20 may determine the location and posture of the portable three-dimensional image measuring device 100 to which the marker 310 is attached, based on the location and posture of the marker 310.

[0092] In operation 515, the external electronic device 20 according to various embodiments may perform image matching between the three-dimensional image of the surface of the object and a medical image of the object. For example, the external electronic device 20 may transform

the coordinate system of the three-dimensional image of the surface of the object into the coordinate system of the external electronic device 20 by mediation of the marker 310 attached to the portable three-dimensional image measuring device 100. The external electronic device 20 may transform the coordinate system of the medical image into the coordinate system of the external electronic device 20. After completing the coordinate system transformation, the external electronic device 20 may perform image matching between the three-dimensional image of the surface of the object and the medical image of the object.

[0093] FIG. 6 is an operation flowchart of the portable three-dimensional image measuring device 100 according to various embodiments of the present disclosure.

[0094] Referring to the operation flowchart 600, in operation 610, the portable three-dimensional image measuring device 100 according to various embodiments may irradiate an object with patterned light output by the light source 120. For example, the portable three-dimensional image measuring device 100 may output the patterned light through the light source 120. The output patterned light may be reflected by the optical path control element 140 and the object may be irradiated with the reflected patterned light.

[0095] In operation 620, the portable three-dimensional image measuring device 100 according to various embodiments may receive reflected light generated by reflecting the patterned light from the object. For example, the patterned light with which the object is irradiated may be reflected by the object and incident again into the portable three-dimensional image measuring device 100. The reflected light may pass through the optical path control element 140 and reach the camera 130.

[0096] In operation 630, the portable three-dimensional image measuring device 100 according to various embodiments may generate a light field image of the object by receiving the reflected light through the camera 130. In operation 640, the portable three-dimensional image measuring device 100 according to various embodiments may generate a three-dimensional image of the surface of the object based on the light field image of the object. In operation 650, the portable three-dimensional image measuring device 100 according to various embodiments may transmit the three-dimensional image of the surface of the object to the external electronic device.

[0097] FIG. 7 is a view showing an example in which the medical image matching system 10 according to various embodiments of the present disclosure is used.

[0098] Referring to FIG. 7, a doctor (D) may acquire a three-dimensional image of a surface of a patient (P) using the portable three-dimensional image measuring device 100. For example, the doctor D may use the portable three-dimensional image measuring device 100 to irradiate the surface of the patient P with patterned light. A pattern 710 may be formed on the surface of the patient P by the patterned light with which the surface of the patient P is irradiated. The portable three-dimensional image measuring device 100 may generate a light field image of the patient P by receiving reflected light reflected from the patient P. The light field image of the patient P may be, for example, an image in which a plurality of sub-images of the irradiated pattern 710 are combined. The portable three-dimensional image measuring device 100 may generate the three-dimensional image of the surface of the patient P using the light field image of the patient P. The portable three-dimensional

image measuring device **100** may transmit the generated three-dimensional image of the surface of the patient **P** to the external electronic device **20**.

[0099] The external electronic device **20** according to various embodiments may capture an image of at least a portion of the pattern surface of the marker **310** attached to the portable three-dimensional image measuring device **100** through an imaging device to form a pattern image of the at least a portion of the pattern surface. The external electronic device **20** may determine the location and posture of the portable three-dimensional image measuring device **100** to which the marker **310** is attached, based on the formed pattern image.

[0100] The external electronic device **20** according to various embodiments may transform or align the coordinate system of the three-dimensional image of a surface of the patient **P** into the coordinate system of the external electronic device **20**. For example, the external electronic device **20** may transform the coordinate system of the three-dimensional image of the surface of the patient **P** into the coordinate system of the external electronic device **20** based on the location and posture of the portable three-dimensional image measuring device **100** determined through the marker **310**.

[0101] The external electronic device **20** according to various embodiments may transform or align the coordinate system of a medical image of the patient **P**, which is received from a medical device, to the coordinate system of the external electronic device **20**. The external electronic device **20** according to various embodiments may perform image matching by unifying the coordinate systems between the three-dimensional image of the surface of the patient **P** and the medical image of the patient **P**.

[0102] FIG. **8** is a view for explaining a structure of the camera **130** according to various embodiments of the present disclosure.

[0103] Referring to FIG. **8**, the camera **130** may include a condensing lens **137**, a lens array **135**, and an image sensor **131** sequentially disposed from an object **810**. The camera **130** is an example of an arrangement structure for acquiring a light field image, but it may, of course, have a structure different from the depicted structure for acquiring the light field image.

[0104] The condensing lens **137** according to various embodiments is a component that condenses reflected light reflected from the object **810**. The condensing lens **137** may be a convex lens with one focal length so that the reflected light reflected from the object **810** is condensed to one point. When the condensing lens **137** is implemented using a plurality of lenses, etc., the plurality of lenses may be defined as one thin lens according to a known thin lens theory. Accordingly, the diameter, focal length, and center of the condensing lens **137** may be respectively expressed as the diameter, focal length, and center of one thin lens defined in this way.

[0105] The lens array **135** according to various embodiments may disperse light entering through the condensing lens **137** and condense the light to a plurality of points formed at different locations. The lens array may be composed of a plurality of micro lenses. For example, the lens array **135** may be disposed closer to the condensing lens **137** than the focal length of the condensing lens **137**. For

example, the lens array **135** may be disposed farther from the condensing lens **137** than the focal length of the condensing lens **137**.

[0106] The lens array **135** according to various embodiments may be disposed at a location corresponding to the focal length of the condensing lens **137**. In this case, the focus of the light entering from the condensing lens **137** may be formed on one of a plurality of micro lenses **135a**. In addition, the image sensor **131** may be fixedly installed at a location corresponding to the focal length of each micro lens **135a** included in the lens array **135**.

[0107] The image sensor **131** according to various embodiments may sense the light passing through the lens array **135**. In addition, the image sensor **131** may acquire a light field image including a plurality of sub-images corresponding to the plurality of points. The image sensor **131** may include at least one imaging element of any type configured to acquire an imaging image of any object, and the image sensor **131** may be composed of a plurality of pixels **131a**.

[0108] The image sensor **131** according to various embodiments may output, for example, a light field image having a format of photo aggregate file when photographing once. The photo aggregate file may include a plurality of sub-images having depths of different subjects in which the focal points of the object are formed at locations corresponding to the focal points of the plurality of micro lenses. Both color information and direction information of light may be stored in each sub-image according to X and Y coordinates.

[0109] The sub-images according to various embodiments may have different subject depths but may photograph the same object. The shapes of the object seen in the sub-images may be substantially the same, and a difference may occur between the locations of a clearly visible portion and a blurry portion. The clearly visible portion may be a portion where the focus of the corresponding micro lens **135a** is formed and has the subject depth, and the blurry portion may be a portion excluding this.

[0110] The light field camera according to various embodiments may be configured to determine a subject depth posteriorly after photographing the object and combine images having different subject depths. Accordingly, the image sensor of the light field camera may have a posterior and variable subject depth. In addition, the light field image generated by the light field camera may include a plurality of sub-images that store both color information and direction information of light.

[0111] In another embodiment, the camera **130** may perform a refocusing process using a plurality of sub-images. In the refocusing process, an image having a desired depth may be newly extracted by combining a desired subject depth and color information of pixels corresponding to the optical path or direction calculated backward accordingly, among pixels of the light field image. Through this, it is possible to generate an image in which the irradiated pattern can be clearly identified.

[0112] FIG. **9** is a view for explaining the lens array **135** of the camera **130** according to various embodiments of the present disclosure.

[0113] According to various embodiments, the plurality of micro lenses **135a** included in the lens array **135** may be provided in N numbers (where N is a natural number of 1 or more). That is, N may mean a plurality. For example, in the lens array **135**, i micro lenses may be disposed in each row

and j micro lenses may be disposed in each column. Accordingly, N micro lenses may be composed of $i*j$ matrices. For example, in order to form a more compact light field, the lens array **135** may have a shape in which approximately 1,000*1,000 micro lenses are arranged. The arrangement and number of micro lenses may vary according to various conditions (e.g., physical properties, photographing environment, required resolution of sub-images, or the number of pixels of an image sensor) of the condensing lens **137** and the micro lenses.

[0114] The plurality of N micro lenses according to various embodiments may disperse light entering through the condensing lens **137** into N points. The image sensor **131** shown in FIG. **8** may be divided into N areas corresponding to the N points formed by the N micro lenses. The focal points of the N micro lenses may be formed to be dispersed into the N areas of the image sensor **131**.

[0115] According to various embodiments, when N sub-images are formed in the N areas, the light field image may include N sub-images having different subject depths. In addition, the processor may select an image having a subject depth at a predetermined location among the N images.

[0116] FIG. **10** is a view showing a process of forming different subject depths of a plurality of sub-images included in the light field image acquired by the camera **130** according to various embodiments of the present disclosure.

[0117] The camera **130** according to various embodiments may include a condensing lens **137**, a lens array **135**, and an image sensor **131**. A first object **1010** may be disposed closer to the condensing lens **137** than a second object **1020**.

[0118] According to various embodiments, light emitted from the top of the first object **1010** may be condensed through the condensing lens **137** and focused on a micro lens **135c** disposed on the lower side of the lens array **135**. Light emitted from the micro lens **135c** may reach an area **A1** disposed on the lower side of the image sensor **131**. Accordingly, the amount of light emitted from the top of the first object **1010** may be mainly distributed in the lower area **A1** and a small amount of light may be distributed in the other areas. That is, an image of the top of the first object **1010** may be clearly formed on pixels **131e**, **131f**, and **131g** included in the lower area **A1**.

[0119] According to various embodiments, light emitted from the top of the second object **1020** may be condensed through the condensing lens **137** and focused on the micro lens **135b** in the middle of the lens array **135**. Light emitted from the micro lens **135b** may reach an area **A2** disposed in the middle of the image sensor **131**. Accordingly, the amount of light emitted from the top of the second object **1020** may be mainly distributed in the middle area **A2** and a small amount of light may be distributed in the other areas. That is, an image of the top of the second object **1020** may be clearly formed on pixels **131b**, **131c**, and **131d** included in the middle area **A2**.

[0120] Since a small amount of light emitted from the second object **1020** is distributed in the lower area **A1**, an image of the second object **1020** may be formed in a blurry state. In addition, since a small amount of light emitted from the first object **1010** is distributed in the middle area **A2**, an image of the first object **1010** may be formed in a blurry state. Accordingly, the lower area **A1** may output a sub-image with a subject depth for the first object **1010**, and the middle area **A2** may output a sub-image with a subject depth for the second object **1020**.

[0121] As described above, when an object is photographed by a camera according to various embodiments, a light field image including a plurality of sub-images having different subject depths may be generated.

[0122] FIG. **11** is a view for explaining a light field image **1100** including a plurality of sub-images having different subject depths according to various embodiments of the present disclosure.

[0123] Referring to FIG. **11**, the light field image **1100** according to various embodiments may be output as a photo aggregate file including a plurality of sub-images in which subject depths are formed in areas **C1**, **C2**, **C3**, and **C4** at different locations in the image sensor **131**. The plurality of areas **C1**, **C2**, **C3**, and **C4** may be located at different locations within the sub-images, and depending on circumstances, at least two areas may be located at the same location. The photo aggregate file may have an aggregate format in which a plurality of sub-images physically separated from each other are simply collected. Unlike this, the photo aggregate file may be a format in which a plurality of sub-images are integrally combined with each other using a new extension method. According to various embodiments, the sub-images may include color information and direction information of light to have different subject depths. A direction of an arrow shown in FIG. **11** may represent a direction in which a distance at which a subject depth is to be formed increases.

[0124] While the foregoing methods have been described with respect to particular embodiments, these methods may also be implemented as computer-readable codes on a computer-readable recording medium. The computer-readable recording medium includes any kind of data storage devices that can be read by a computer system. Examples of the computer-readable recording medium includes ROM, RAM, CD-ROM, magnetic tape, floppy disk, optical data storage device and the like. Also, the computer-readable recording medium can be distributed to the computer systems which are connected through a network so that the computer-readable codes can be stored and executed in a distributed manner. Further, the functional programs, codes and code segments for implementing the foregoing embodiments can easily be inferred by programmers in the art to which the present disclosure pertains.

[0125] Although the technical spirit of the present disclosure has been described by the examples described in some embodiments and illustrated in the accompanying drawings, it should be noted that various substitutions, modifications, and changes can be made without departing from the scope of the present disclosure which can be understood by those skilled in the art to which the present disclosure pertains. In addition, it should be noted that such substitutions, modifications and changes are intended to fall within the scope of the appended claims.

What is claimed is:

1. A portable three-dimensional image measuring device comprising:

- a light source configured to output patterned light;
- a camera configured to generate an image of an object by receiving reflected light generated by reflecting the patterned light from the object;
- an optical path control element configured to reflect the patterned light output from the light source so that the object is irradiated with the patterned light, and to

transmit the reflected light reflected from the object so that the reflected light reaches the camera; and
 a processor,
 wherein an optical path of the patterned light output from the light source and irradiated onto the object and an optical path of the reflected light reflected from the object and reaching the camera overlap coaxially in a section between the optical path control element and the object, and
 wherein the processor is configured to:
 measure a light intensity of reflected light at each of a plurality of points constituting a surface of the object using the image of the object acquired through the camera;
 generate phase data based on the measured light intensity; and
 generate a three-dimensional image of the surface of the object by calculating a height of each of the plurality of points based on the generated phase data.

2. The portable three-dimensional image measuring device of claim 1, wherein the image is a light field image in which a plurality of sub-images color information and direction information of the reflected light are combined.

3. The portable three-dimensional image measuring device of claim 1, further comprising a communication circuit,
 wherein the processor is configured to transmit the three-dimensional image of the surface of the object to an external electronic device through the communication circuit.

4. The portable three-dimensional image measuring device of claim 1, further comprising a marker installed to be movable from a predetermined location of the portable three-dimensional image measuring device,
 wherein the processor is configured to transmit information indicating displacement of the marker from the predetermined location to an external electronic device.

5. The portable three-dimensional image measuring device of claim 1, wherein the light source and the camera are disposed in a direction perpendicular to each other based on the optical path control element.

6. The portable three-dimensional image measuring device of claim 1, wherein the camera includes:
 a lens array in which a plurality of micro lenses are arranged; and
 an image sensor that captures the reflected light passed through the lens array.

7. The portable three-dimensional image measuring device of claim 1, wherein the light source includes:
 a pattern generator in which a plurality of patterns are formed; and
 an LED that irradiates the pattern unit with light.

8. The portable three-dimensional image measuring device of claim 7, wherein the light output by the LED is infrared light.

9. The portable three-dimensional image measuring device of claim 1, wherein the optical path control element is a transmissive mirror.

10. The portable three-dimensional image measuring device of claim 1, further comprising:
 a first housing in which the light source, the camera, and the optical path control element are disposed; and

a second housing coupled to the first housing and formed with an opening through which the object is irradiated with the patterned light output from the light source.

11. A three-dimensional image measuring method of a portable three-dimensional image measuring device, comprising:
 irradiating an object with patterned light output by a light source through an optical path control element; and
 generating an image of the object by receiving reflected light, which is generated by reflecting the patterned light from the object, by a camera through the optical path control element;
 measuring a light intensity of reflected light at each of a plurality of points constituting a surface of the object using the image of the object acquired through the camera;
 generating phase data based on the measured light intensity; and
 generating a three-dimensional image of the surface of the object by calculating a height of each of the plurality of points based on the generated phase data,
 wherein an optical path of the patterned light output from the light source and irradiated onto the object and an optical path of the reflected light reflected from the object and reaching the camera overlap coaxially in a section between the optical path control element and the object.

12. The three-dimensional image measuring method of claim 11, wherein the image is a light field image in which a plurality of sub-images color information and direction information of the reflected light are combined.

13. The three-dimensional image measuring method of claim 11, further comprising:
 transmitting the three-dimensional image of the surface of the object to an external electronic device.

14. The three-dimensional image measuring method of claim 11, further comprising:
 transmitting information indicating displacement of a marker, which is installed to be movable from a predetermined location of the portable three-dimensional image measuring device, from the predetermined location to an external electronic device.

15. The three-dimensional image measuring method of claim 11, wherein the light source and the camera are disposed in a direction perpendicular to each other based on the optical path control element.

16. The three-dimensional image measuring method of claim 11, wherein the camera includes:
 a lens array in which a plurality of micro lenses are arranged; and
 an image sensor that captures the reflected light passed through the lens array.

17. The three-dimensional image measuring method of claim 11, wherein the light source includes:
 a pattern generator in which a plurality of patterns are formed; and
 an LED that irradiates the pattern unit with light.

18. The three-dimensional image measuring method of claim 17, wherein the light output by the LED is infrared light.

19. The three-dimensional image measuring method of claim 11, wherein the optical path control element is a transmissive mirror.

20. The three-dimensional image measuring method of claim **11**,

wherein the light source, the camera, and the optical path control element are disposed in a first housing, and wherein a second housing is coupled to the first housing and formed with an opening through which the object is irradiated with the patterned light output from the light source.

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